

# WORLD HABITATS

**Seen from space, the most striking feature** of the earth is its sheer physical variety. Not only is there dry land and open sea, but there are also mountains, plains, rivers, coastal shelves, and deep oceanic trenches. The earth also varies in its climate: in some parts of the world, weeks or months pass by under almost cloudless skies, while in others, the ground is scoured by icy winds or soaked by intense tropical storms. Differences like these create a complex jigsaw puzzle of varied habitats, enabling the Earth to support a rich diversity of animal life. Some species are highly adaptable, and can survive in a wide range of conditions, but the vast majority are found in one kind of habitat and nowhere else.

## What are habitats?

In its narrowest sense, a habitat is the environment in which something lives. For some animals, a habitat might be as restricted as a temporary pool in a desert or as small as a piece of decaying wood. In a broader sense—the one used in this book—a habitat can mean a characteristic grouping of living things, together with the setting in which they are found. In ecology, a habitat defined in this way is known as a biome.

Habitats contain both living and nonliving matter. In some—for example, true desert—living things are thinly scattered, so the nonliving part of the environment is dominant. In others, such as forest and coral reefs, living things are so abundant that they fill all the available space and create habitats for each other. In these habitats, huge numbers of species exist side by side, forming extraordinarily complex webs of life.

## Factors that shape habitats

Geology plays a part in shaping habitats, but by far the most important factor is climate. As a result, differences in climate—which sometimes occur over remarkably small distances—can have a huge effect on plant and animal life. A classic example of this occurs where mountain ranges intercept rain-bearing winds. On the windward side of the mountains, heavy rainfall often creates lush forests teeming with all kinds of animal life. But in the



### EFFECTS OF CLIMATE

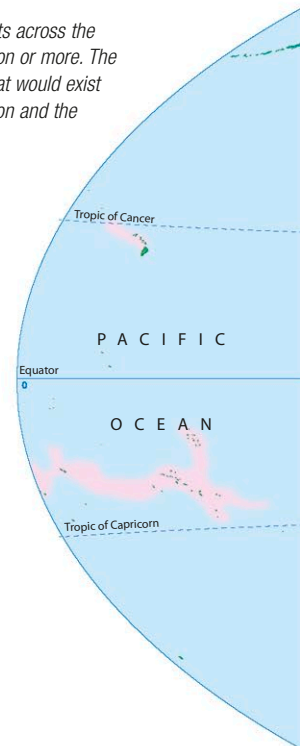
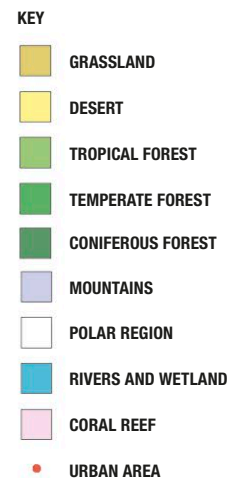
*These two habitats—in Argentina (left) and Chile (right)—are at the same latitude, but they have very different climates, and therefore very different plant and animal life. The contrast is created by the Andes, a mountain barrier that blocks rain-bearing winds. Chile is on the windward side of the Andes.*

“rainshadow,” to the lee of the mountains, low rainfall can produce desert or scrub, where only drought-tolerant animals can survive.

Temperature is another climatic factor that has an important effect both on land and in the sea. For example, in the far north, coniferous forest eventually peters out in the face of biting winter frosts. This northern tree line, which runs like a ragged ring around the Arctic, marks the outer range of crossbills, wood wasps, and many other animals that depend on conifers for survival. On coasts and at sea, temperature

### HABITATS OF THE WORLD

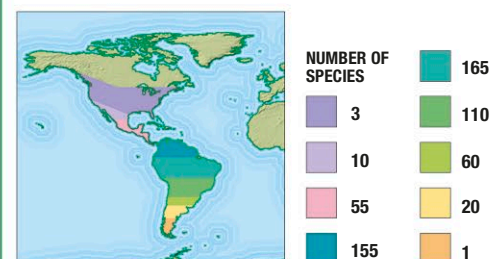
*This map shows the distribution of major habitats across the world, and also cities with populations of 5 million or more. The habitat distribution shown here is the pattern that would exist if man-made changes, resulting from urbanization and the spread of agriculture, had not occurred.*



changes are usually more gradual than they are inland. However, warmth—or lack of it—still determines where some habitats are found. For example, reef-building corals do not thrive at temperatures of less than about 68°F (20°C), so most reefs are found in the tropics. However, on the west coast of Africa and the Americas, reefs are rare because, although the climate is warm, cold currents pass close to the shore. Mangrove swamps present a similar pattern: in the Southern Hemisphere, they reach as far as South Australia; in the Northern Hemisphere, they extend only just out of the tropics.

## Biodiversity

From the earliest days of scientific exploration, naturalists noticed great variations in biodiversity, or species richness. In the far north and south, species totals are low compared with the numbers found near the equator. Arctic tundra, for example, is inhabited by just a few hundred species of insects, while in tropical forests, the total is probably at least a million. A similar picture—albeit on a smaller scale—is true for mammals and birds. However, high-latitude habitats make up for their lack of biodiversity by having some phenomenally large species populations. The seas around Antarctica, for example, possibly harbor tens of millions of crabeater seals—one of the most numerous wild large mammals on earth.



### TROPICAL TREASURY

*The hummingbird family vividly demonstrates the high level of species diversity that is to be found in the tropics. Only a handful of hummingbird species live at high latitudes—and most of these are migrants—but on the equator, the number of species rises to over 150.*

## CHEMICAL CYCLES

In all habitats, living things take part in cycles that shuttle chemical elements between living and nonliving matter. About 25 elements are essential to life and, of these, just 4 make up the bulk of living things. These are hydrogen, oxygen, nitrogen, and—the key element—carbon. In the nonliving world, carbon can be found in the atmosphere (as a gas), in water (in dissolved form), and in the ground (in rocks and fossil fuels). Plants absorb carbon dioxide from the atmosphere, and most other forms of life give off carbon dioxide when they break down carbon-containing substances to release energy. Carbon is also released by burning fossil fuels.

### THE CARBON CYCLE

*This diagram shows some of the main pathways in the carbon cycle. The time taken for each part of the cycle to be completed varies greatly. Carbon may stay in living things for only a few days, but it can remain locked up underground for thousands of years.*

